GridIQ

Site assessment: overview

Demo Corporation · Durban Distribution Ctr

Assessment
GQ-A-0004 · 7-day assessment

Report
revision 2 · issued

Observation period (SAST)
2026-11-02 21:00 -> 2026-11-09 21:00

Observed
7.00 days of 7

Site address
88 Logistics Blvd, Durban

Measurement points
PV (pv), Warehouse Main (incomer)

This revision replaces revision 1: layout fix: blank footer pages removed from the PDF

1. Executive findings

- Investigate (PV, confidence high; for: GridIQ): Only 11 % of the expected measurements were received; conclusions for this point are limited.
- Investigate (Warehouse Main, confidence medium; for: electrical contractor): Current unbalance reached 30.41 % (95th percentile, unbalance as computed by the meter (per its configuration); not stated to be negative-sequence unbalance). Rebalancing single-phase loads usually helps.

2. Installation and measurement boundary

Point	Role	Nominal	CT primary	Contracted MDI	Device · installed (SAST)
PV	pv	230 V	300 A	500 kVA	device · 2026-11-05 22:17
Warehouse Main	incomer	230 V	–	500 kVA	device · 2026-08-05 22:45

Point	Coverage	Samples	Reporting interval	Gaps >= 30 min	Meter
PV	10.9 %	13 131	5 s	3	device
Warehouse Main	100.0 %	14 003	300 s	2	device

Coverage = samples received × the meter's median reporting interval ÷ the observation window. Missing data is never filled or shown as zero.

sum of 1 top-level incomer(s); sub-meters are inside these and are not added

3. Energy, demand and operating pattern

PV

Energy imported
4 073,5 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
177,2 kVA at 2026-11-06 15:30

Contracted MDI
500 kVA (35 % used)

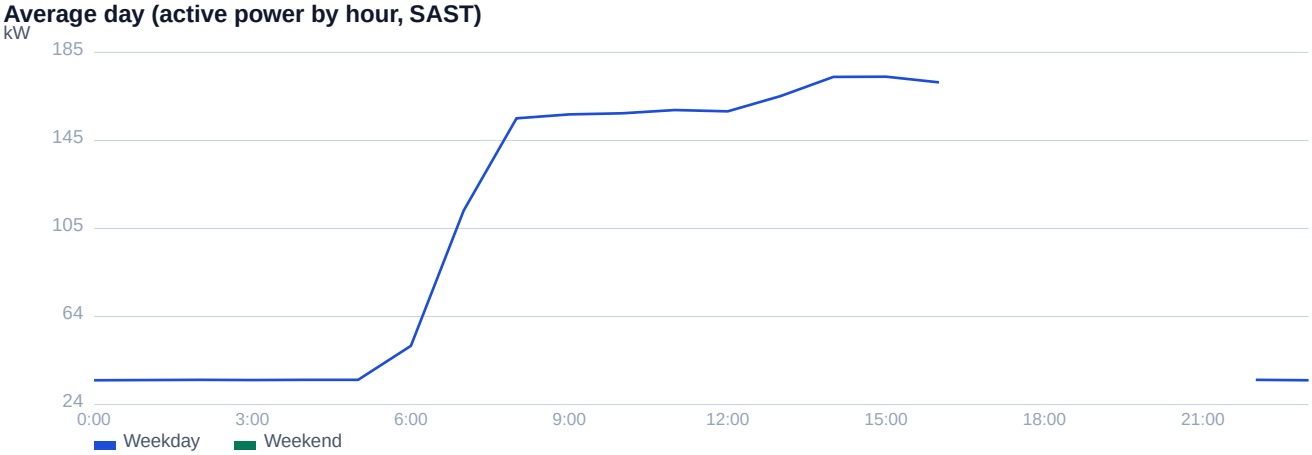
Highest instantaneous active power
180,5 kW (minute of 2026-11-06 15:36)

Baseload (10th percentile)
34,9 kW

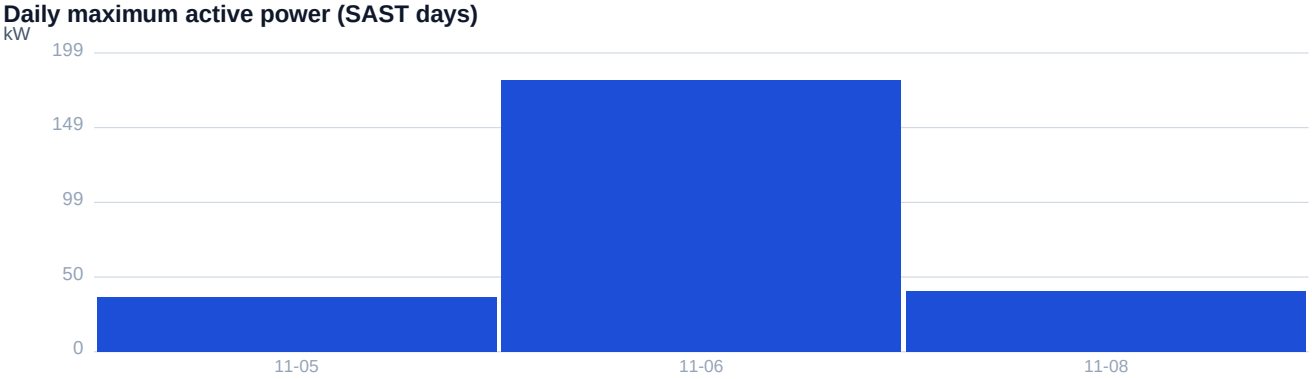
Operating-hours average
148,7 kW

Outside operating hours
35,6 kW

Average day (active power by hour, SAST)



Daily maximum active power (SAST days)



hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

Warehouse Main

Energy imported
29 455,6 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
403,6 kVA at 2026-11-03 13:00

Contracted MDI
500 kVA (81 % used)

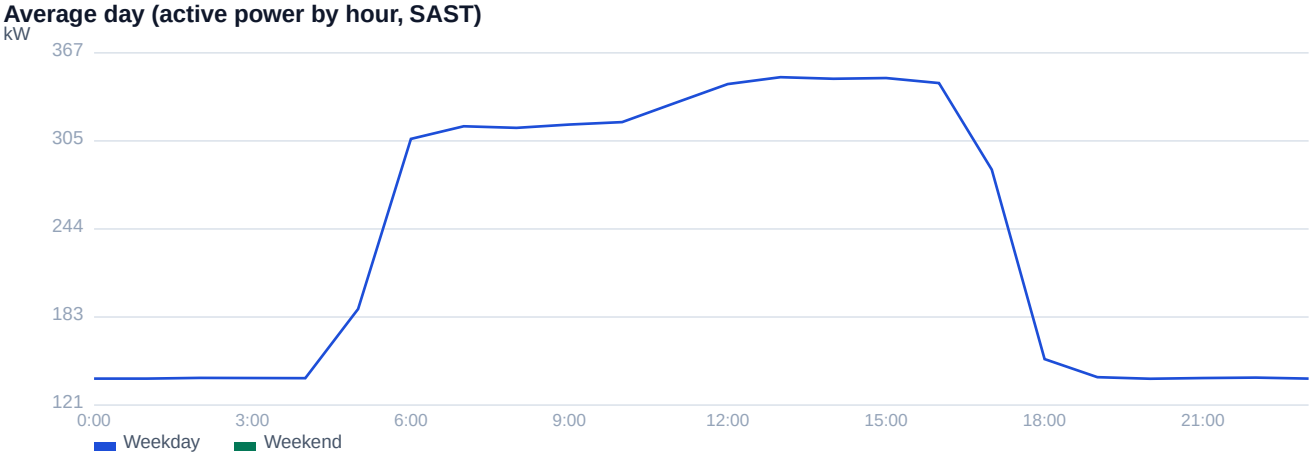
Highest instantaneous active power
382,1 kW (minute of 2026-11-03 12:15)

Baseload (10th percentile)
139,1 kW

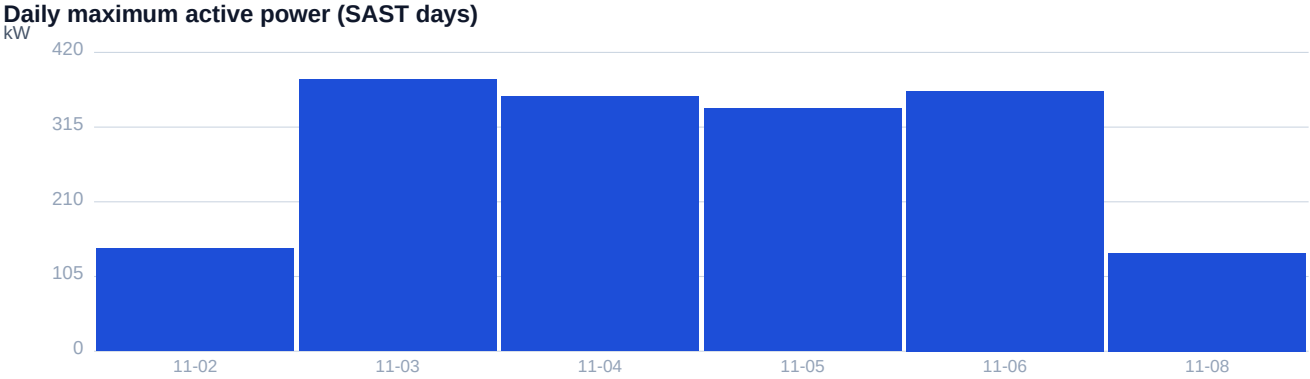
Operating-hours average
327,2 kW

Outside operating hours
144,6 kW

Average day (active power by hour, SAST)



Daily maximum active power (SAST days)



hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

4. Electrical summary

- PV: voltage A 226,3–234,8 V, B 226,3–234 V, C 227–235,6 V; THD-V p95 A 2.5 %, B 2.5 %, C 2.6 %; PF (loaded) 0.972; 0 minute(s) below 23 V.
- Warehouse Main: voltage A 220,3–229,2 V, B 223,4–232,3 V, C 226,2–234,5 V; THD-V p95 A 3.2 %, B 3.1 %, C 3.2 %; PF (loaded) 0.920; 0 minute(s) below 23 V.

Screening against NRS 048-2 thresholds; see the power-quality report for detail and the basis of every figure.

5. Notable events and data limitations

- Observation period 7 days (2026-11-02T19:00:36.240Z to 2026-11-09T19:00:36.240Z); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.
- PV: 11 % of expected measurements received (3 gap(s) of 30 min or more).
- PV: the meter's capability profile is not confirmed; only channels present in the data are reported.
- Warehouse Main: the meter's capability profile is not confirmed; only channels present in the data are reported.

0 voltage/frequency excursion event(s) recorded; inferred from the sampled RMS trend (pq_events), not device-recorded.

6. Prioritised actions

#	Action	Who investigates	Urgency	Confidence
1	Only 11 % of the expected measurements were received; conclusions for this point are limited.	GridIQ	next maintenance	high
2	Current unbalance reached 30.41 % (95th percentile, unbalance as computed by the meter (per its configuration); not stated to be negative-sequence unbalance). Rebalancing single-phase loads usually helps.	electrical contractor	next maintenance	medium

7. Feasibility-data readiness

Supplied: 30-minute interval load profile (kW, kVA); daily energy; billed maximum demand with timestamps; power-quality screening results; metering topology and observation limits.

Missing inputs: declared operating hours (default 06:00–18:00 weekdays used); supply details and tariff; existing PV / generator / battery details; fault level at the point of supply; a full month or more of data (seasonal and monthly patterns).

GridIQ supplies measured data; system sizing and financial modelling belong to the feasibility study.

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
79b49ad7529769dca21be0162d67ec10…

Calculation version
gridiq-calc-2026.11.1

Rule version
gridiq-rules-2026.11.1

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-09 21:00

Generated (SAST)
2026-11-11 21:13

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 79b49ad7529769dca21be0162d67ec106d55c4fa8e310cfa3036c896474837e7